DriveNow Database Design Documentation

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PAGE 3 – QUESTIONS & ASSUMPTIONS

1. Questions

ID	SRS Section	Question
Q1	7.1	Can a branch manager manage a different branch over time, and should the system preserve all previous manager assignments?
Q2	7.1	When a regional director temporarily replaces a branch manager, should this temporary assignment be recorded in the same way as a normal manager assignment?
Q3	7.1	Can an employee belong to more than one branch, or must every employee have exactly one assigned branch?
Q4	7.2	Can an employee enroll in the same course multiple times, and should every attempt have its own attempt number?
Q5	7.3	How should the system identify a vehicle with an individual price that differs from its category's daily rate?
Q6	7.3, 7.6	When a vehicle is transferred, should its home branch change, or should the system keep the original home branch and record its current physical location separately?
Q7	7.5	Can a GPS tracking device be moved from one vehicle to another, and should the system preserve its previous installation history?
Q8	7.7	What happens to authorized drivers and their historical rental records when a corporate account is closed?
Q9	7.8	If two corporate discounts apply to the same rental category and period, which discount should take priority?
Q10	7.10	How are loyalty points calculated for each completed rental, and how many points are required to redeem a discount?
Q11	7.11	Can a reservation be modified after confirmation, and which changes should be recorded in its history?
Q12	7.12	Can a rental be returned to a branch different from the planned return branch?
Q13	7.12	How is the additional daily fee for each additional driver calculated and recorded?
Q14	7.16	How are corporate monthly invoices linked to multiple rentals and their payments?

2. Assumptions

ID	SRS Section	Design Assumption
A1	7.1	Each employee belongs to exactly one branch at a time.
A2	7.1	A branch manager is also stored as an employee.
A3	7.1	Previous branch manager assignments must be preserved using assignment history records.
A4	7.2	Each course attempt is identified by EmployeeID, CourseCode, and AttemptNo.

ID	SRS Section	Design Assumption
A5	7.3, 7.6	A vehicle's home branch and its current physical branch are stored separately.
A6	7.3	Historical category prices are stored with their validity periods and are never overwritten.
A7	7.4	The quantity and actual daily charge of each extra are stored for each rental.
A8	7.5	A GPS device can be uninstalled and installed in another vehicle, so installation history should be preserved.
A9	7.7	Individual and corporate customers are mutually exclusive customer types.
A10	7.7	Authorized driver numbers are unique within each corporate customer, not across all companies.
A11	7.10	Loyalty points are calculated from recorded point movements rather than stored as an independently editable balance.
A12	7.12	A rental can exist without a reservation because walk-in rentals are allowed.
A13	7.12	The receiving agent is nullable until the vehicle is returned.
A14	7.13	The price charged for each insurance package is stored in the rental record to preserve historical pricing.
A15	7.16	Corporate monthly invoices are modeled separately from individual rental payments.

3. Contradictions

No explicit contradiction was identified in the reviewed SRS statements. The following items are unresolved design questions, not confirmed contradictions:

SRS Sections	Observation
7.12 and 7.16	Section 7.12 describes payments associated with individual rentals, while Section 7.16 says corporate clients usually receive one invoice at the end of each month. This requires clarification of the corporate billing model; it is not necessarily a contradiction.
7.3 and 7.6	Section 7.3 identifies a vehicle's home branch, while Section 7.6 describes transfers between branches. The design must distinguish the responsible home branch from the vehicle's current physical location.

PAGE 4 – BUSINESS RULES

These rules require procedural or application-level enforcement because the listed declarative constraints alone cannot fully enforce the required cross-record checks or automatically maintain an audit history.

BR1 – Business Rule

SRS Section	Business Rule
7.7	A blacklisted customer cannot create a new reservation.

Why the listed constraints are not enough
The system must check the customer's blacklist status in another record before allowing a reservation. A simple CHECK constraint cannot validate this condition across related records.

Enforcement Method
Trigger or Application Logic: Before creating a reservation, check whether the customer has an active blacklist record. Reject the reservation if the customer is blacklisted.

BR2 – Business Rule

SRS Section	Business Rule
7.12	For a corporate rental, the main driver must be one of the authorized drivers belonging to that same corporate customer.

Why the listed constraints are not enough
The system must verify that the selected driver belongs to the same company associated with the rental. This requires checking related records and the rental's customer type.

Enforcement Method
Stored Procedure or Application Logic: Before creating the rental, verify that the selected driver is registered as an authorized driver for the rental's corporate customer. Reject invalid assignments.

BR3 – Business Rule

SRS Section	Business Rule
5 (O-3), 7.3	Every important vehicle status change must record who made the change and when it happened.

Why the listed constraints are not enough
A normal CHECK or FK constraint cannot automatically preserve a historical record of every status change.

Enforcement Method
Database Trigger: After every vehicle status update, insert the old status, new status, employee ID, and change timestamp into a vehicle status history table.

Design note

BR3 is linked to Business Objective O-3 in Section 5 and the vehicle status tracking requirement in Section 7.3. The proposed trigger requires a vehicle status history table containing at least the vehicle, old status, new status, employee, and change timestamp.

Review note: Questions and assumptions are proposed for design review. Confirm them with the business analyst before treating them as approved requirements. Do not record unresolved questions as SRS contradictions unless two SRS statements explicitly conflict.